



Developing and validating evacuation models for fire safety engineering

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ABSTRACT

Evacuation models can adopt different approaches for the simulation of human behaviour in fire. This paper provides an overview of the most commonly used modelling methods to represent the evacuation process in a fire scenario. This is presented through a structure matching the engineering time-line model of evacuation. The evacuation model development process is discussed considering both data-driven empirical correlations as well as theory-based modelling approaches. Examples of alternative methods to the currently used evacuation modelling assumptions are also presented. These methods have been chosen to provide examples of cases in which revisions of well-established assumptions may be needed. This review mainly focuses on buildings and pedestrian evacuation scenarios. Nevertheless, many concepts presented are potentially applicable to traffic evacuation. Particular attention is given to the representation of the impact of smoke on the evacuation process, as this is an important issue for fire safety engineering. Finally, a discussion on existing methods and procedures for the verification and validation of evacuation models is presented and the need for their standardization is advocated.

1. Introduction

Evacuation models are used in fire safety engineering (FSE) to investigate the safety conditions of a given building, area or infrastructure. Their use has been increasing over the years due to the steady growth of countries allowing performance-based design [1] and the need to design buildings for rapid evacuation [2]. Similarly, evacuation models may be a useful tool for enhancing emergency preparedness and management of large outdoor fire incidents [3]. The simulation of human behaviour in fire poses several challenges from a model development perspective and requires a multi-disciplinary approach to the subject. Several disciplines play a fundamental role in evacuation model development, including engineering, psychology, applied mathematics, physics, computer science, sociology, etc. While an individual researcher might not have a comprehensive expertise in all these domains, in recent years, several cross-disciplinary efforts have been conducted to bridge the gaps between disciplines and improve the validity of human behaviour predictions in case of fire evacuation [4].

To date, the pedestrian evacuation modelling market for FSE applications include over 70 models [5]. While the first evacuation models can be dated as early as the 70ies [6], the key assumptions adopted by evacuation models adopted today in FSE have mostly been developed in the late 80ies and 90ies. Examples of such assumptions include the representation of the space using a fine network or a grid [7] or a

continuous two-dimensional space [8] connected by vertical elements (i.e., stairs or elevators) or inclined planes (i.e. ramps). A similar observation can be made for the sub-models adopted for the simulation of pedestrian movement and navigation, e.g. the social force model [9] or the steering model [10], which are generally applied in two-dimensional spaces. This means that today, despite three-dimensional graphics showing otherwise, to the author's knowledge, there is no fully 3D evacuation simulation of people movement. In addition, apart from a few exceptions [11], most evacuation models represent exit/route choice using very simple assumptions (e.g., shortest or quickest path algorithms) [12]. These methods can only partially take into account the complexity of human decision making, the full set of personal/environmental factors and the heuristic processes that people undertake while deciding their route/path [13]. Their basic mathematical assumptions may need to be revised as they are based on optimization techniques/algorithms. In contrast, several studies demonstrated that humans tend to "satisfy rather than optimize" in emergency situations [14], thus indicating that people would not tend to necessarily adopt the quickest or shortest path in an actual emergency.

The results of a recent survey [5] identifying the most used pedestrian evacuation models in FSE further demonstrate the mismatch between the most known/used evacuation models and recent research developments. None of the top three most used models identified adopt a movement modelling method developed in the last twenty years (i.e.,

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they implement the steering model [15], a cellular automata approach [16] and the social force model [17]).

This analysis poses an outstanding question: why have the modelling approaches developed over the past twenty years not become mainstream in pedestrian evacuation models for FSE applications? On one hand, **model development takes time and any new modelling approach requires time before becoming a well-established, validated and accepted method for commercial applications**. On the other hand, model developers might have felt the need to concentrate their efforts on other issues, such as expanding the capability of an existing model, increasing the number of sub-models included and/or behavioural aspects considered, collecting more data for validation, improving graphics of result visualization, etc., rather than questioning and/or modifying the core underlying assumptions in use. **Another perspective could be that existing models have been successfully employed for FSE applications, thus a need for a paradigm shift in their basic assumptions might not be needed.**

While the great bulk of evacuation research in the FSE community has been focusing on pedestrian evacuation, a few studies have addressed evacuation in case of large outdoor fires (e.g. fires at the wildland-urban interface, city fires, etc.). These scenarios require the simulation of both human response as well as pedestrian and traffic movement [18]. A review [19] identified 22 traffic models which could potentially be used for such scenarios and it highlighted that most of these models have not been specifically designed for FSE applications. Therefore, their assumptions and integration with fire models need to be further developed [20]. It should be noted that although this paper mostly focuses on pedestrian evacuation models, many of the concepts presented here are applicable to traffic evacuation models as well.

Given these premises, this paper has a dual aim. Firstly, it wants to provide an overview of the most used modelling approaches adopted by evacuation models for FSE applications. Secondly, it is hoped that it will stimulate a debate on if/which fundamental modelling assumptions should be questioned. To achieve this, examples of alternative approaches for the representation of human behaviour compared to what are today “mainstream” approaches are reviewed and discussed. Those examples should by no means be considered exhaustive and are mostly drawn from the literature of the author of this paper and his network of collaborators. Their objective is not merely to advocate for their use, but rather to generate a broad discussion among the fire evacuation model developers and users on the key assumptions that should be revised and why. In this context, a broad analysis about the methods used for the validation of evacuation model results is presented, given its importance for assessing the reliability of model predictions. Finally, this paper suggests the development of a common standard for evacuation model validation to reduce inconsistencies among modelling results.

2. The evacuation model development process

Considering a model as a simplified representation of reality [21], different approaches can be used for the development of its main assumptions. The first approach resembles the traditional Galilean scientific method in which a theory/modelling method concerning the representation of human behaviour in fire is first formulated and then analysed/validated against evacuation data-sets. An alternative approach is to provide predictions of human behaviour with data-driven empirical models. The latter models do not require a full understanding of the system modelled, but rather aim at using existing data-sets for the identification of trends/behaviours that can be predicted. In other words, the model developers rely on an appropriate choice of statistical model fitting to interpret a given data-set on evacuation movement and provide predictions accordingly [22]. Both approaches present advantages and limitations, but they rely on a reasonable underlying premise, i.e., the results produced have to be based on/checked with data. In this context, it is necessary to review the assumptions of evacuation models that may be driven by the simplicity of implementation rather than data

or theories [23]. The actions (or inactions) simulated by evacuation models are generally represented following the engineering time-line model [24] and these are here presented considering the specific needs of model applications in the FSE domain.

3. Modelling human behaviour using a time-line

The time-line model is a simplified method to represent the phases of an evacuation. Two different time-lines can be used to represent the evacuation process, namely an engineering and behavioural timeline (see Fig. 1) [25].

It should be noted that this time-line approach should be applied for each individual evacuee. A behavioural time-line can be used to explain the (in)actions observed during evacuation. This includes detection and alarm (starting from ignition until deviation from normal occurs), the recognition phase (time from deviation from normal until a decision to act to reduce consequences is taken) and then response (which includes both non-purposive and purposive movement towards a safe place). The engineering time-line includes the awareness phase (corresponding to detection and alarm in the behavioural time-line), the pre-movement phase (time from deviation from normal until purposive evacuation movement) and the movement phase (time to reach a safe place once people have started moving) [25]. The engineering time-line is currently adopted in evacuation models, thus this is used as structure for reporting the modelling approaches. The awareness phase (including detection and alarm times) is generally represented implicitly in evacuation models within the pre-movement time or it is neglected (see red lines in Fig. 1) [26]. For this reason, the following sections focus only on the pre-movement and movement phase.

3.1. The pre-movement phase

The term *pre-movement* is here used although several other terms can be found in the literature to refer to this stage (e.g., pre-travel activities, pre-evacuation, delay, etc.). The word pre-movement is often criticized as it seems to refer to a time in which movement does not occur, while authors argue that people do not remain in place in this phase [27]. Considering the engineering time-line, the term pre-movement should therefore be intended as the time before purposive movement towards a safe place may occur.

The most common approach to represent the pre-movement phase is the use of pseudo-random sampling techniques to assign values from distributions to represent the variability of human decisions [28]. Despite several studies recommending the use of log-normal distributions for this phase [24], evacuation models generally provide the opportunity to choose among different types of distributions. Along with log-normal distributions (given to provide a non-symmetric distribution), sigmoid curves are often available in traffic evacuation models [29], while pedestrian evacuation models may also include other distributions (e.g., uniform, normal, triangular, exponential, Weibull, etc.). The user would have to reflect on the most appropriate distribution to use, trying to identify their match with actual data [22]. This approach is deemed to represent implicitly the factors affecting both the time for the recognition of a fire emergency as well as human response time. Nevertheless, in most evacuation models, this results in agents remaining stationary in their initial location until their pre-movement time is over [30], a behaviour which does not seem to realistically represent the actual human course of actions. Evacuation models may also give the opportunity to explicitly represent the activities that might occur during this phase through behavioural itineraries. In this case, the user can represent each action, its duration and its specific purpose. The key issue with these approaches is that the represented behaviour relies on user's assumptions rather than an actual model prediction [31].

An alternative approach is the use of a predictive sub-model of pre-movement which explicitly considers the factors affecting evacuation decision. A possible modelling approach of this type has been proposed

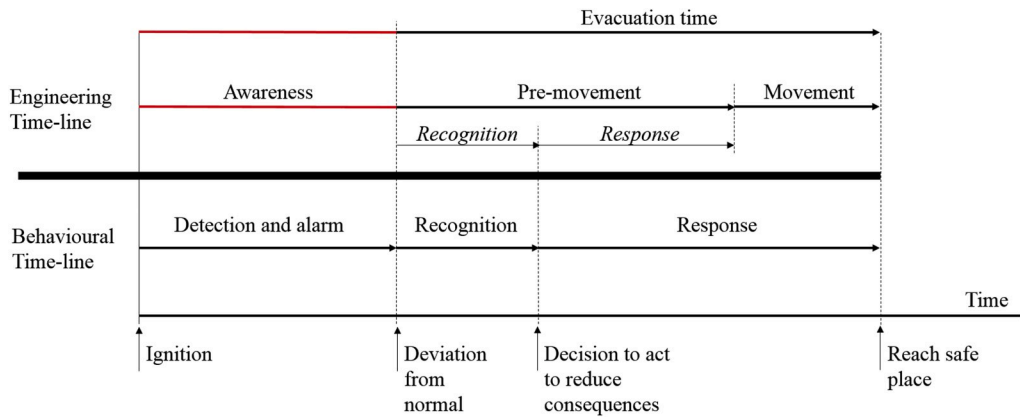


Fig. 1. The engineering and behavioural time-lines used to represent evacuation. Re-drawn from Ref. [25].

in the literature and it relies on the representation of key internal and external factors (e.g., risk perception [32]) using random utility theory [30,33]. The main advantage of this approach is that the user would not be requested to provide a distribution *a priori*, as the model would represent the influence of the factors influencing evacuation decision. Such type of model is therefore more in line with existing behavioural theories describing the pre-movement phase [34] and would leave less room to uncertainty due to the user effect [35]. Its main limitation is the need for a large number of dedicated data-set(s) for calibration. Despite the scarcity of data concerning pre-movement behaviour being a known issue in the field [36,37], it is expected that an increased knowledge in human behaviour in fire would lead to the development of more predictive models for pre-movement decision.

3.2. The movement phase

The movement phase is represented in evacuation models at different levels, mainly considering 1) tactical/strategic/global level (i.e. route/exit choice), and 2) local level (e.g. representing local movement and collision avoidance with obstacles/other agents) [27]. An additional locomotion level at a more refined degree of resolution may be considered (intended here as the detailed representation of the components of an agent, e.g., legs or shoulders of a pedestrian or the wheels of a car), although this is often neglected in evacuation models [27]. In the context of FSE, fire smoke can have a significant impact on evacuation movement, thus this aspect has been addressed in this paper in a dedicated sub-section.

3.2.1. Route/exit choice modelling

The methods employed to represent route/exit choice can be grouped into four main categories, namely 1) shortest path, 2) quickest path, 3) conditional and 4) user-defined. Most evacuation models are based on optimization algorithms aimed at obtaining paths which resemble the shortest (i.e., reaching the closest exit) or the quickest path (i.e., reaching the exit associated with the shortest time). More sophisticated approaches include the representation of conditions which may affect the chosen route and the availability of exits, e.g., lines of sight, the presence of smoke, social influence, etc. [11]. Experimental studies [38] have shown that environmental conditions (e.g. smoke) can affect human navigation choices, e.g., people may decide to walk closer to the walls in low visibility conditions. Similarly, several studies have demonstrated that way-finding systems can have a significant impact on the chosen evacuation routes [39–42]. Evacuation models generally allow their representation in an implicit way, by letting the user define custom routes (i.e. similarly to the previously mentioned concept of behavioural itineraries used in the pre-movement phase) or assign the agents to specific exits. The use of specified routes can also be useful to model assisted evacuation scenarios, in which a given itinerary could be

assigned to an agent in order to represent the process of providing assistance to evacuees that are not able to proceed with self-evacuation [43].

Considering an example of a microscopic agent-based model, the choice of an individual agent in a simple room with two exits can be modelled in several manners. Fig. 2 presents a set of simulated paths considering examples of shortest (see Fig. 2.1), quickest (see Fig. 2.2) and user-defined (see Fig. 2.6) paths along with three examples of conditions which may be taken into consideration, e.g., an exit unavailable from the beginning of the simulation (see Fig. 2.3), an exit unavailable during the simulation, considering information provided by a signage system (see Fig. 2.4) and a simulated “follow the leader” behaviour to account for social influence (see Fig. 2.5) [44]. This shows that varying levels of sophistication can be adopted to represent route choice in which different factors can be implicitly or explicitly taken into consideration. Assessing the impact of different assumptions gets more difficult with the increasing complexity of a building. For instance, route choices in a multi-storey building include both horizontal and vertical egress components (e.g., stairs, elevators) which might lead to several combinations of chosen routes/exits. The complexity of route choice can subsequently impact evacuation design [45].

While the most sophisticated pathfinding approaches [46] implement criteria for dynamic re-routing and an exploration of the space during navigation, most evacuation models assume the so called “god’s view”, i.e. the simulated agents are fully aware of the space from the beginning of the simulation and they generally have an initial target to reach. This approach can be considered applicable for evacuation scenarios in which people are fully familiar with the environment. Nevertheless, this assumption should be scrutinized in transient scenarios where the evacuees are not fully aware of the geometric layout of the building, particularly for buildings in which spatial navigation is complex. Given the fact that evacuation models are generally used for performance-based design in the FSE domain, it is expected that a complex geometric layout might be commonplace rather than an exception (as prescriptive-based design is generally adopted for simpler buildings). Developing and testing appropriate criteria for dynamic re-routing could significantly improve the predictive capabilities of evacuation models and possibly provide a more reliable assessment of the actual routes used by evacuees [11]. In addition, it appears evident the need to further investigate how people perceive the surroundings and other people (in)actions and respond to the cues along their evacuation path to further develop route choice sub-models.

3.2.2. Movement modelling

The representation of evacuation movement is generally performed considering movement on a horizontal plane and then connecting these spaces with vertical egress components. In this way, agents move in a three-dimensional space (although the actual movement is generally

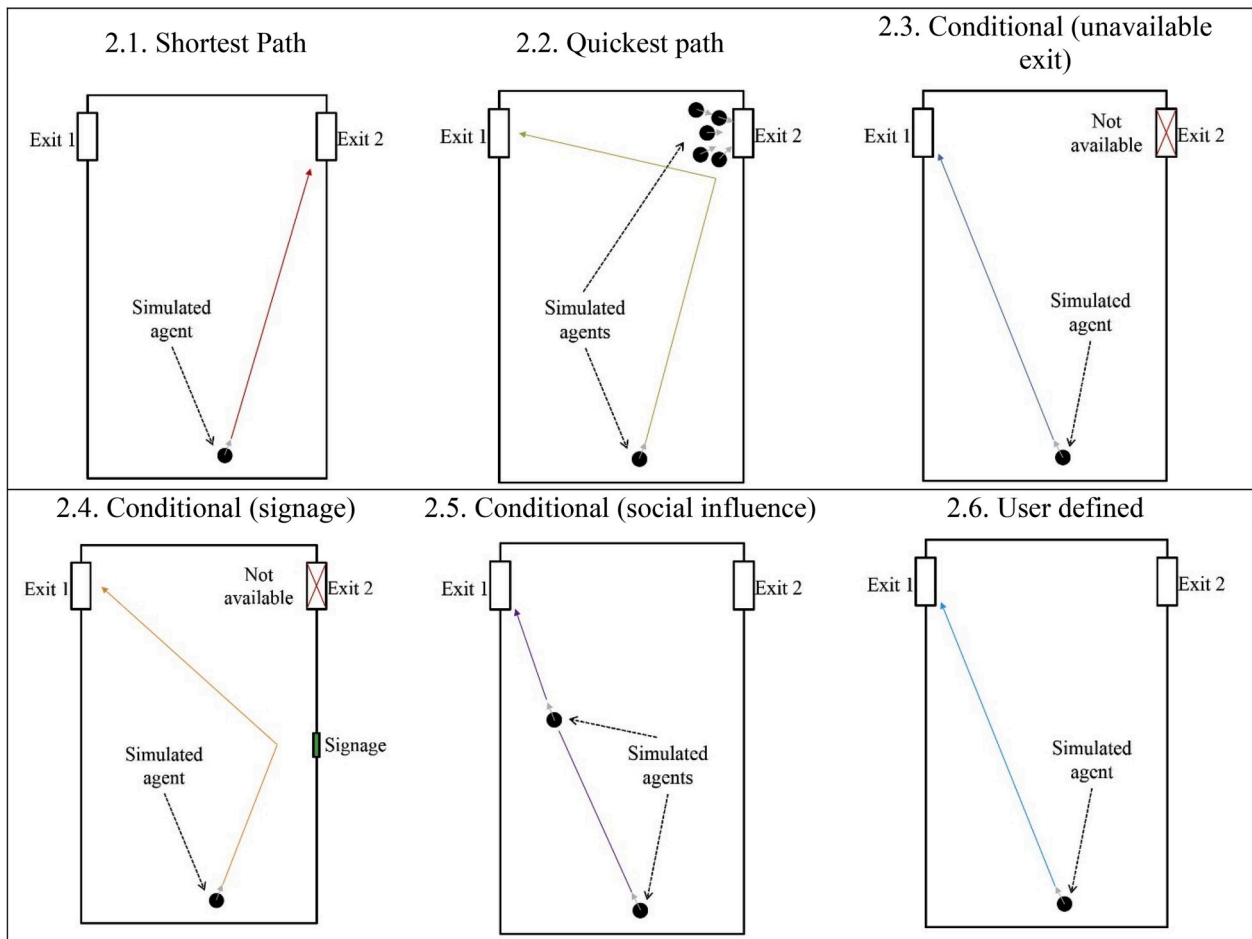


Fig. 2. Example of chosen routes based on the assumptions adopted for a simple choice between two exits. [2.1]: Path based on shortest distance; [2.2]: Path based on quickest time (in this case a queue is present at the nearest exit); [2.3]: Path based on a condition (the agent knows one exit is not available from the beginning of the simulation); [2.4]: Path based on a condition (the agent knows one exit is not available once receiving the information from a sign); [2.5]: Path based on a condition (the agent selects one exit based on the choice of another agent); [2.6]: The exit is deterministically assigned by the user.

calculated in two-dimensions). A common method in use is the adoption of empirical relationships between speed, density and flows [47]. A set of equations are therefore implemented in either a discrete or continuous space and movement of the agents is determined by these relationships.

Cellular automata (CA) can be employed to represent evacuation movement. It should be noted that the term CA is not used here solely to define the discretization of space (i.e., the use of a fine network modelling approach). The term CA here indicates that each cell contains a discrete set of assigned variables that can be used to represent people movement. In fact, the space in CA models consists of a uniform lattice made of cells, which can be either occupied (by an agent or an obstacle) or available. A time-based or event-based approach can be used to update the state of the cells and the movement takes place in accordance to a given set of rules (the use of floor fields being one of the most common approaches [48]). The main advantages of CA models rely on the simplicity of implementation and reduced computational resources required (mostly because these models may not require the integration of differential equations of movement) [49]. Key limitations are linked to one of the core assumptions of CA models, i.e., all cells have the same size and they can generally be occupied by one agent at a time. This means that results might be sensitive to the grid adopted (given the discrete approximation of space and people movement), it is not possible to represent agents of different dimensions (e.g. an adult and a child would be assumed to have the same size) and the calculated densities depend on the chosen grid rather than actually being predicted [27]. The

latter issue can have significant implications on the reliability of evacuation model predictions as it may affect the resulting relationships between speed, density and flows [47].

The movement of the agents can also be obtained as an emerging property based on a set of global/local rules (i.e. using an agent-based modelling approach) or differential equations (generally based on Newtonian forces) defined by the model developers. Both approaches allow the representation of the interactions between agents and between agents and the space. Examples of such methods for the representation of local movement and collision avoidance are the *social force model* [9] and the *steering model* [10]. In such models, users generally provide a set of distributions of *unimpeded walking speeds* (with the most common being uniform or normal distributions), from which values are sampled and assigned to each agent. Each distribution generally corresponds to a type of population, giving the model user the opportunity to represent explicitly the impact of physical abilities of different populations (e.g. adult, elderly, children, etc. [50–52]). Speeds are then updated at each time-step by the model to calculate the *impeded speeds*.

The unimpeded walking speed is often called *desired walking speed*, as it indicates the speed that agents would choose in absence of external interference. In other words, models assume that agents have a maximum achievable speed which is then reduced by the presence of other people or obstacles. This assumption is inspired by *self-driven particle models* [53], which assume that agents are self-propelled particles with a constant speed which is aligned to the one of their neighbours and consider a random perturbation. This assumption was originally

designed for models representing movement of particles from an origin to a destination. While the use of a constant value of desired/unimpeded speed is a well-established assumption in the evacuation modelling literature, it implies that the desired speed is not depending on variable internal/external factors. Experimental studies have shown that the desired walking speeds are affected by the perceived urgency of the situation [54]. An alternative approach for representing the desired speed would therefore consider a variable speed. An example of a conceptual model that represents explicitly this issue argues that the desired walking speed should not merely depend on the physical abilities of the evacuees, but it should be linked to his/her motivation to move along with physical abilities [55]. This could be linked to factors such as physical exertion, mental fatigue and risk perception [32,55]. This type of movement models would allow the attainment of desired walking speeds that are updated over time, and they may correspond to a more accurate representation of movement. This approach could also be adopted to represent additional behavioural factors affecting the selection of the desired walking speed (e.g. social influence, group behaviours, etc. [56]).

In recent years, the movement on vertical egress components has received much lower attention if compared with horizontal movement [57], while early studies put great emphasis on stair evacuation movement [58]. Concerning people movement on stairs, evacuation models either adopt similar assumptions as if the movement would happen on an incline or a projected plane (this is used for instance in cellular automata approaches [49]) or make use of empirical equations derived from stair movement data-sets [59,60]. Of particular interest is the case of representing merging flows coming from different egress components (e.g. stairs and floors), as they can affect the order in which floors are evacuated [61]. This issue is particularly important in phased evacuation strategies as their successful outcome may depend on the clearing time of each floor. Dedicated models for the study of the impact of merging flows on the evacuation process are available in the literature [62] and it is hoped that future model developments would further investigate this issue.

Fire regulatory frameworks have also introduced the possibility to adopt vertical means of egress other than stairs especially for particularly tall buildings and deep underground facilities [45]. The most used alternative to stairs in case of fire emergencies is the use of Occupant Evacuation Elevators (OEE). A set of evacuation models have therefore incorporated dedicated sub-models for the representation of OEE and the decision making process required to choose between vertical egress components [57,63–65]. As much of the data and modelling strategies are generally considering healthy young populations, the use of OEE can be a possible solution to address the issues of people with disabilities [64].

3.3. Modelling the impact of smoke

A set of evacuation models gives the option to represent explicitly the impact of smoke on evacuation [66]. As the environmental conditions are evolving due to the fire, both the pre-movement and evacuation phases can be affected. For instance, fire cues and smoke can be used to influence the simulated decision to evacuate [67]. Smoke can also affect the movement phase, by impacting route selection [11] or walking speeds [66]. Several experimental research studies have demonstrated that walking speed is affected by reduced visibility conditions [66], with a trend of reduction of speed with increasing extinction coefficient (see Fig. 3). Limited research is available on other impacts of fire smoke, such as irritancy or toxicity [66]. Different empirical correlations have been built and implemented in evacuation models to represent the impact of smoke, based on different interpretations of the existing data-sets [68,69].

It should be noted that most evacuation models available on the market allow only for a one-way coupling between fire simulations and evacuation behaviour [70]. This means that the fire evolution can affect

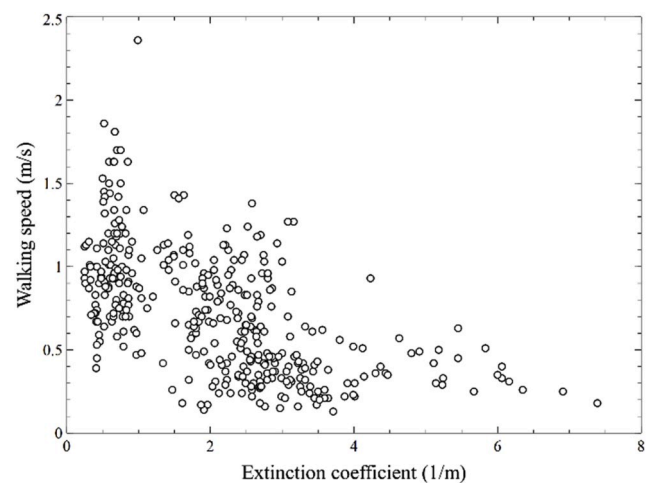


Fig. 3. Aggregated experimental data-sets [66] concerning walking speeds in relation to extinction coefficient.

the simulated human behaviour, but evacuees' actions (e.g. fire extinguishing attempts, opening doors, etc.) cannot automatically impact the fire.

4. Verification and validation of evacuation models

Two surveys among model users showed that verification and validation (V&V) are the main contributing factors for selecting an evacuation model [5,71]. Verification is here intended as the *process of determining that a calculation method implementation accurately represents the developer's conceptual description of the calculation method and the solution to the calculation method* [72]. Validation is here defined as the *process of determining the degree to which a calculation method is an accurate representation of the real world from the perspective of the intended uses of the calculation method* [72]. Many methods and procedures are today available to evaluate the predictive capabilities of egress models [73]. Table 1 presents a summary of the tests included in the three main existing V&V documents (this is further discussed in Ref. [74]), along with a short description of the component they evaluate.

One of the most used documents on evacuation model V&V has been published by the International Maritime Organization (IMO). IMO has provided a list of tests for the V&V of evacuation model capabilities within a set of guidelines on the assessment of new and existing passenger ships. The guidelines were presented first within the MSC/Circ. 1238 [75], and later revised within the MSC/Circ. 1533 [76]. Among the topics discussed in these guidelines, a key objective is the assessment of the ability of evacuation models to reproduce a range of key egress components (defined as *component testing*) along with checking that the model has the range of capabilities required for the intended simulations. A qualitative analysis of the predicted human behaviour is also recommended; these guidelines also state that there is no reliable data available to perform this analysis quantitatively [76].

Another V&V document available in the literature consists of a set of guidelines provided by the German group RiMEA [*Richtlinie für Mikroskopische EntfluchtungsAnalysen*, corresponding to Guidelines for microscopic analysis of evacuation] [77]. The objective of this document was to provide a methodology for the development and testing of evacuation models, including setting minimum standards to be met by such models, as well as a list of tests of key components to be analysed. Similarly to the IMO guidelines, the tester is required to present the full range of capabilities of the model under consideration and provide instructions for its correct use. As in the IMO guidelines, a qualitative analysis of the model predictions is recommended, while a quantitative analysis is considered not yet possible due to the lack of reliable experimental data.

Table 1

Summary of the verification tests presented within the IMO [76], RiMEA [77] and NIST [78] documents. The word “no” indicates that a given component is not evaluated with a dedicated test in the corresponding document.

Tested component	IMO	RiMEA	NIST
Pre-movement time assignment	Test 5	Test 5	Test.1.1
Walking speed in a corridor	Test 1	Test 1	Test 2.1
Walking speed on stairs	Tests 2 & 3	Tests 2 & 3	Test 2.2
Movement around a corner	Test 6	Tests 6 & 15	Test 2.3
Assigned demographics	Test 7	Tests 7 & 8	Test 2.4
Horizontal counter-flows	Test 8	No	Test 2.8
People with movement disabilities	No	No	Test 2.10
Group behaviour	No	No	Test 2.9
Reduced visibility vs walking speed	No	No	Test 2.5
Occupant incapacitation	No	No	Test 2.6
Lift usage	No	No	Test 2.7
Exit route allocation	Test 10	Test 10	Test 3.1
Dynamic availability of exit	No	No	Test 4.1
Social Influence on exit choice	No	No	Test 3.2
Affiliation to familiar exits	No	No	Test 3.3
Route choice	No	Tests 11 & 14	No
Congestion in front of a flight of stairs	Test 11	Test 13	Test 5.1
Maximum flows	Test 4	Test 12	Test 5.2
Stair flow rates	No	No	No
Relationship between walking speed, uni-directional flows and densities	Test 12	Test 4	No
Crowd of people leaving a large public space	Test 9	Test 9	No

Following this issue, a document from the National Institute of Standards and Technology (NIST) [78,79] reviewed the existing tests available in the literature and provided a modified list of tests for building fire evacuation models. Tests were grouped based on the main core behavioural components they belong to (i.e. pre-movement, movement and navigation, route/exit usage, route availability and selection and flow constraints). The document also provided references to data-sets which have been considered potentially suitable for validation.

An additional key aspect to consider when evaluating the predictive capabilities of evacuation models is the assessment of the uncertainties associated with the results produced. As most evacuation models make use of a probabilistic approach [80], it is crucial to assess the convergence of results [81]. This is often linked to the so-called *behavioural uncertainty*, i.e. the uncertainty linked with the use of pseudo-random sampling from distributions causing that the same input configuration can provide different results [28,82]. A few methods have been proposed in the literature to address this issue, including different statistical and mathematical techniques, such as functional analysis [28], statistical inference testing [83], and confidence intervals [84].

5. Discussion

A common application of evacuation models in the FSE domain is performance-based design. Nevertheless, many models and sub-models used in FSE have been originally designed for other applications [23]. For instance, most pedestrian movement models have been developed to investigate crowd dynamics. Similarly, many traffic models have been designed for optimization of traffic flows rather than assessing evacuation scenarios [19]. For this reason, an important step for the evacuation modelling community working in the FSE domain is to evaluate if the assumptions behind these models can be reasonably applicable to FSE. This paper is not necessarily advocating for a drastic change in the core assumptions adopted by evacuation models, but it rather wants to encourage a debate on reviewing their suitability for FSE applications.

This paper is primarily targeting model developers, as the overview of modelling methods presented here along with examples of alternative

approaches may generate useful ideas for future model improvements. Nevertheless, it is also deemed to be relevant to other groups. Researchers can identify a list of challenges with the currently available assumptions adopted by evacuation models and pursue research efforts aimed at addressing them. Regulators might consider this paper as a useful read to identify solutions aimed at testing more rigorously the validity of the results obtained with evacuation models. Model users might identify strengths and weaknesses of existing evacuation models and take counter-measures in their engineering practice.

While this paper attempted to provide an exhausting list of core assumptions adopted by evacuation models, there are definitely several other aspects which can play a role in evacuation model development, validation and applications. For example, one aspect which is not explicitly considered in evacuation models is the impact of evacuees' experience on evacuation decisions. In particular, repeated unnecessary evacuations could potentially affect the decision to evacuate and the pre-movement time, possibly decreasing compliance to an evacuation order or an alarm. This issue is often referred in the literature as the *cry wolf effect* [85]. A recent modelling attempt to address this issue has been performed adopting game theory [85], mostly due to its unique ability to investigate the interactions between decision makers. This example can be used to perform a general reflection on the current use of evacuation models. In general terms, evacuation models are generally addressing a punctual evacuation scenario in a given condition of the building and with a given (set of) population(s). This might be seen as a rational way to address the problem as in most cases, buildings will be affected by, at most, one major fire during their lifetime. An alternative approach may include the simulation of the lifetime of a building, placing fire evacuation safety in a larger picture; this may allow considering the implications of the occurrence of real fire threats and false alarms (along with other variables) on evacuation decision making. In this context, a fire safety assessment performed within Building Information Modelling (BIM) tools could be a suitable platform to address this issue [86]. This is given the fact that BIM tools are designed to investigate the lifetime of a building and give the opportunity to develop a multi-layer model which can constantly be updated over time.

While reviewing currently used modelling methods versus alternative (often more sophisticated) approaches, an important issue to consider is the trade-off between reductionism and parsimony. This is intended as the need to develop models which are at the same time simple enough to be successfully used by fire engineers without excluding necessary aspects of human behaviour [87]. The future adoption of the approaches discussed in this paper as alternative to mainstream approaches is linked to several factors, including the increased availability of experimental data for their calibration and computational resources. In addition, the level of granularity adopted in the representation of human behaviour is strictly linked to the field of application of evacuation models. This is also related to the fact that model users may require a higher level of sophistication in relation to the aim of their analysis. In this context, this paper wants to encourage a debate on the need for the representation of very detailed behaviour at the individual level versus the use of simplified assumptions. This is linked to the variability of evacuation scenarios, including cases in which the behaviour of one individual can have a significant impact on the outcome.

Concerning the V&V of evacuation models, despite many research efforts that have been conducted over the years to define suitable V&V tests and procedures [17,77,78,88], a standardized document which can be used by all interested parties does not exist yet. This issue has one main consequence on the model users: they need to choose among different evacuation models without the opportunity to review their capabilities in a consistent format. A standardized V&V document cannot be designed by a single researcher, developer or research/development group but it rather requires a broad contribution of the evacuation modelling community. Based on this premise, an international task group has been created within the Fire Safety Committee 92

of the International Standardization Organization (ISO). The aim of this task group is to develop a standardized protocol for the verification and validation of evacuation models used in buildings based on the previous research conducted in the area. The task group involves evacuation modelling experts from the industry, model developers and researchers, including experts that have been involved in the development of some of the tools currently on the market. This document is currently under development and is deemed to represent a significant milestone towards the assessment of the key requirements that evacuation models need to meet in order to be used in fire evacuation design. It is hoped that similar efforts will take place in the future for evacuation models applied in other fields of FSE (e.g., traffic evacuation models).

While a standardized document would provide a useful protocol for the assessment of evacuation model capabilities, several other issues need to be further studied. In particular, from a regulatory perspective, it is crucial to investigate how acceptance criteria should be identified and who should assess model results. In addition, issues concerning the default configuration of evacuation model should also be considered. The simulation results produced by model users are deemed to be affected by the model developer choice of a given default setting, and the degree of conservatism to be included in defaults is still an open issue [35]. It should also be noted that future areas of development of evacuation models might come from other, fast growing, areas of research, such as pattern recognition, machine learning, big data, etc. It is hoped that research advancements in these fields will allow the collection and treatment of more and more amount of data which can be implemented into evacuation models both using data-driven approaches as well as stimulate the development of new human behaviour in fire theories.

6. Conclusion

This paper has provided an overview of the main assumptions and modelling methods adopted in evacuation modelling for FSE. Examples of alternative approaches have been presented and compared to existing solutions adopted in evacuation models. The aim of this paper is to encourage a broad debate on the need for challenging well-established assumptions and to provide a broad review on the modelling solutions adopted to represent the evacuation time-line. The state-of-the-art of the methods used for the verification and validation (V&V) of evacuation models is also presented. Given the inconsistencies among testing methods for V&V, this paper also highlights the need for a standardized protocol to address this issue.

Declaration of competing interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] E. Gissi, E. Ronchi, D.A. Purser, Transparency vs magic numbers: the development of stair design requirements in the Italian Fire Safety Code, *Fire Saf. J.* 91 (Jul. 2017) 882–891.
- [2] E. Ronchi, Disaster management: design buildings for rapid evacuation, *Nature* 528 (7582) (Dec. 2015), 333–333.
- [3] E. Ronchi, G. Rein, S. Gwynne, R. Wadhvani, P. Intini, A. Bergstedt, "e-Sanctuary, Open Multi-Physics Framework for Modelling Wildfire Urban Evacuation," *Fire Protection Research Foundation*, Quincy, MA (USA), 2017.
- [4] M. Kinatader, E. Ronchi, Letter to the editor. Burning Down the Silos in a Multidisciplinary Field. Towards Unified Quality Criteria in Human Behaviour in Fire," *Fire Technol.* May 2019, 10694-019-00867-z.
- [5] R. Lovreglio, E. Ronchi, M.J. Kinsey, An Online Survey of Pedestrian Evacuation Model Usage and Users 56, *Fire Technology*, 2020, pp. 1133–1153. In this issue, <https://doi.org/10.1007/s10694-019-00923-8>.
- [6] V. Bazjanac, Simulation of elevator performance in high-rise buildings under conditions of emergency, *Hum. Response Tall Build.* (1977) 316–328.
- [7] E. Galea, J.P. Galparsoro, Exodus: an evacuation model for mass transport vehicles, Civil Aviation Authority, 1993.
- [8] P.A. Thompson, E.W. Marchant, "Testing and application of the computer model 'SIMULEX', *Fire Saf. J.* 24 (2) (Jan. 1995) 149–166.
- [9] D. Helbing, P. Molnár, Social force model for pedestrian dynamics, *Phys. Rev. E* 51 (5) (May 1995) 4282–4286.
- [10] C.W. Reynolds, "Steering Behaviors for Autonomous Characters," Presented at the Game Developers Conference, 1999, pp. 763–782.
- [11] E. Ronchi, D. Nilsson, S.M.V. Gwynne, Modelling the impact of emergency exit signs in tunnels, *Fire Technol.* 48 (4) (Apr. 2012) 961–988.
- [12] K. Bladström, Route Choice Modelling in Fire Evacuation Simulators, "LUTVDG/TVBB, 2017.
- [13] M. Moussaïd, D. Helbing, G. Theraulaz, How simple rules determine pedestrian behavior and crowd disasters, *Proc. Natl. Acad. Sci. Unit. States Am.* 108 (17) (Apr. 2011) 6884–6888.
- [14] H.A. Simon, Rational choice and the structure of the environment, *Psychol. Rev.* 63 (2) (1956) 129.
- [15] Thunderhead Engineering, "Pathfinder - Technical Reference, 2018.
- [16] Mott MacDonald Simulation Group, "Simulation of Transient Evacuation and Pedestrian MovementS. STEPS User Manual v5.3," 2016.
- [17] Oasys, The Verification and Validation of MassMotion for Evacuation Modelling, "Arup, 2015.
- [18] E. Ronchi, S. Gwynne, in: S.L. Manzello (Ed.), "Computational Evacuation Modeling in Wildfires," in *Encyclopedia Of Wildfires And Wildland-Urban Interface (WUI) Fires*, Springer International Publishing, Cham, 2019, pp. 1–10.
- [19] P. Intini, E. Ronchi, S. Gwynne, A. Pel, "Traffic modeling for wildland-urban interface fire evacuation, *J. Transport. Eng. Part Syst.* 145 (3) (2019), 04019002.
- [20] E. Ronchi, S.M.V. Gwynne, G. Rein, P. Intini, R. Wadhvani, An open multi-physics framework for modelling wildland-urban interface fire evacuations, *Saf. Sci.* 118 (Oct. 2019) 868–880.
- [21] J. Adrian et al., "A glossary for research on human crowd dynamics," *Collect. Dyn.*, vol. 4, Mar. 2019.
- [22] N.W.F. Bode, E. Ronchi, Statistical model fitting and model selection in pedestrian dynamics research, *Collect. Dyn.* 4 (2019) A20.
- [23] E. Ronchi, et al., New approaches to evacuation modelling for fire safety engineering applications, *Fire Saf. J.* 106 (Jun. 2019) 197–209.
- [24] D.A. Purser, M. Bensilum, Quantification of behaviour for engineering design standards and escape time calculations, *Saf. Sci.* 38 (2) (Jul. 2001) 157–182.
- [25] H. Frantzich, D. Nilsson, K. Rød, "Utrymning och tekniska installationer i vägtunnlar med dubbelriktad trafik," *Lund Brand, Lunds Tekn. Högsk.*, 2016.
- [26] S.M.V. Gwynne, D.A. Purser, D.L. Boswell, Pre-Warning Staff Delay: A Forgotten Component in ASET/RSET Calculations, "presented at the Pedestrian and Evacuation Dynamics, Gaithersburg, MD (US), 2011, pp. 243–253.
- [27] E. Ronchi, D. Nilsson, in: A. Cuesta, O. Abreu, D. Alvear (Eds.), "Basic Concepts and Modelling Methods," in *Evacuation Modeling Trends*, Springer International Publishing, Cham, 2016, pp. 1–23.
- [28] E. Ronchi, P.A. Reneke, R.D. Peacock, A method for the analysis of behavioural uncertainty in evacuation modelling, *Fire Technol.* 50 (6) (Nov. 2014) 1545–1571.
- [29] A.J. Pel, M.C.J. Bliemer, S.P. Hoogendoorn, A review on travel behaviour modelling in dynamic traffic simulation models for evacuations, *Transportation* 39 (1) (Jan. 2012) 97–123.
- [30] R. Lovreglio, E. Ronchi, D. Nilsson, A model of the decision-making process during pre-evacuation, *Fire Saf. J.* 78 (Nov. 2015) 168–179.
- [31] E. Kuligowski, Predicting human behavior during fires, *Fire Technol.* 49 (1) (Nov. 2011) 101–120.
- [32] M.T. Kinatader, E.D. Kuligowski, P.K. Reneke, R.D. Peacock, A Review of Risk Perception in Building Fire Evacuation, "National Institute of Standards and Technology, Sep. 2014. NIST TN 1840.
- [33] R. Lovreglio, E. Ronchi, D. Nilsson, An Evacuation Decision Model based on perceived risk, social influence and behavioural uncertainty, *Simulat. Model. Pract. Theor.* 66 (Aug. 2016) 226–242.
- [34] E. D. Kuligowski, S. M. V. Gwynne, M. J. Kinsey, and L. Hulse, "Guidance for the model user on representing human behavior in egress models," *Fire Technol.*, vol. 53, no. 2, pp. 649–672, Mar. 2017.
- [35] S.M.V. Gwynne, E. Kuligowski, M. Spearpoint, E. Ronchi, Bounding defaults in egress models, *Fire Mater.* 39 (4) (Jun. 2015) 335–352.
- [36] R.F. Fahy, G. Proulx, Toward creating a database on delay times to start evacuation and walking speeds for use in evacuation modeling, in: "2nd International Symposium on Human Behaviour in Fire, 2001, pp. 175–183.
- [37] S.M.V. Gwynne, K.E. Boyce, Engineering data, in: M.J. Hurley, D.T. Gottuk, J. R. Hall, K. Harada, E.D. Kuligowski, M. Puchovsky, J.L. Torero, J.M. Watts, C. J. Wieczorek (Eds.), In *SFPE Handbook Of Fire Protection Engineering*, Springer New York, New York, NY, 2016, pp. 2429–2551.

- [38] E. Ronchi, K. Fridolf, H. Frantzich, D. Nilsson, A.L. Walter, H. Modig, A tunnel evacuation experiment on movement speed and exit choice in smoke, *Fire Saf. J.* 97 (Apr. 2018) 126–136.
- [39] E. Ronchi, et al., A virtual reality experiment on flashing lights at emergency exit portals for road tunnel evacuation, *Fire Technol.* 52 (3) (May 2016) 623–647.
- [40] J. Olander, E. Ronchi, R. Lovreglio, and D. Nilsson, “Dissuasive exit signage for building fire evacuation,” *Appl. Ergon.*, vol. 59, pp. 84–93, Mar. 2017.
- [41] E.R. Galea, H. Xie, S. Deere, D. Cooney, L. Filippidis, An International Survey and Full-Scale Evacuation Trial Demonstrating the Effectiveness of the Active Dynamic Signage System Concept: Evacuation Trials Examining Active Dynamic Signage Systems, “*Fire Mater.*”, 2016.
- [42] S. Arias, S. La Mendola, J. Wahlqvist, O. Rios, D. Nilsson, E. Ronchi, Virtual Reality Evacuation Experiments on Way-Finding Systems for the Future Circular Collider, “*Fire Technol.*”, May 2019.
- [43] V. Alonso-Gutierrez, E. Ronchi, “The Simulation of Assisted Evacuation in Hospitals,” Presented at the Fire and Evacuation Modelling Technical Conference, Malaga, Spain, 2016.
- [44] M. Haghami, M. Sarvi, “‘Herding’ in direction choice-making during collective escape of crowds: how likely is it and what moderates it? *Saf. Sci.* 115 (2019) 362–375.
- [45] E. Ronchi, D. Nilsson, Fire evacuation in high-rise buildings: a review of human behaviour and modelling research, *Fire Sci. Rev.* 2 (1) (2013) 7.
- [46] Xiaoshan Pan, Computational Modeling of Human Behavior for Emergency Egress Analysis: a Multi-Agent Based Simulation Approach, Lambert Academic Publishing, Saarbrücken, 2009.
- [47] A. Seyfried, A. Schadschneider, Fundamental diagram and validation of crowd models, in: H. Umeo, S. Morishita, K. Nishinari, T. Komatsuzaki, S. Bandini (Eds.), *In Cellular Automata*, vol. 5191, Springer Berlin Heidelberg, Berlin, Heidelberg, 2008, pp. 563–566.
- [48] R. Lovreglio, E. Ronchi, D. Nilsson, Calibrating floor field cellular automaton models for pedestrian dynamics by using likelihood function optimization, *Phys. Stat. Mech. Its Appl.* 438 (Nov. 2015) 308–320.
- [49] N. Pelechano, A. Malkawi, Evacuation simulation models: challenges in modeling high rise building evacuation with cellular automata approaches, *Autom. Construct.* 17 (4) (May 2008) 377–385.
- [50] H. Najmanova, E. Ronchi, An experimental data-set on pre-school children evacuation, *Fire Technol.* 53 (4) (Jul. 2017) 1509–1533.
- [51] L. Folk, J. Gales, M. Kinsey, Evacuation simulation of the elderly: data collection and model validation. In *Pedestrian and Evacuation Dynamics Conference Google Scholar*, 2016.
- [52] A. Cuesta, E. Ronchi, S.M.V. Gwynne, “Collection and Use of Data from School Egress Trials,” Presented at the 6th Human Behaviour in Fire Symposium 2015, Downing College, Cambridge, UK, 2015, pp. 233–244.
- [53] T. Vicsek, A. Czirók, E. Ben-Jacob, I. Cohen, O. Shochet, Novel type of phase transition in a system of self-driven particles, *Phys. Rev. Lett.* 75 (6) (Aug. 1995) 1226–1229.
- [54] E. Graat, C. Midden, and P. Bockholts, “Complex evacuation; effects of motivation level and slope of stairs on emergency egress time in a sports stadium,” *Saf. Sci.*, vol. 31, no. 2, pp. 127–141, Mar. 1999.
- [55] E. Ronchi, P.A. Reneke, R.D. Peacock, A conceptual fatigue-motivation model to represent pedestrian movement during stair evacuation, *Appl. Math. Model.* 40 (7–8) (Apr. 2016) 4380–4396.
- [56] M. Moussaïd, N. Perozo, S. Garnier, D. Helbing, G. Theraulaz, The walking behaviour of pedestrian social groups and its impact on crowd dynamics, *PLoS One* 5 (4) (Apr. 2010), e10047.
- [57] M.J. Kinsey, Vertical Transport Evacuation Modelling, “Greenwich, London, UK, 2011.
- [58] J.L. Pauls, Building evacuation: research findings and recommendations. In *Fires and Human Behaviour*, John Wiley & Sons., Canter, D., 1980, pp. 251–275.
- [59] E. D. Kuligowski et al., “Movement on Stairs during Building Evacuations,” National Institute of Standards and Technology, NIST TN 1839, Sep. 2014.
- [60] S.M.V. Gwynne, E.R. Rosenbaum, in: M.J. Hurley, D.T. Gottuk, J.R. Hall, K. Harada, E.D. Kuligowski, M. Puchovsky, J.L. Torero, J.M. Watts, C.J. Wieczorek (Eds.), “Employing the Hydraulic Model in Assessing Emergency Movement,” in *SPPE Handbook Of Fire Protection Engineering*, Springer New York, New York, NY, 2016, pp. 2115–2151.
- [61] T. Sano, E. Ronchi, Y. Minegishi, D. Nilsson, Modelling pedestrian merging in stair evacuation in multi-purpose buildings, *Simulat. Model. Pract. Theor.* 85 (Jun. 2018) 80–94.
- [62] T. Sano, E. Ronchi, Y. Minegishi, D. Nilsson, A pedestrian merging flow model for stair evacuation, *Fire Saf. J.* 89 (Apr. 2017) 77–89.
- [63] E. Ronchi, D. Nilsson, Modelling total evacuation strategies for high-rise buildings, *Build. Simul.* 7 (1) (Feb. 2014) 73–87.
- [64] M. Kinateter, H. Omori, E.D. Kuligowski, The Use of Elevators for Evacuation in Fire Emergencies in International Buildings, National Institute of Standards and Technology, Gaithersburg, MD (US), 2014. Technical Note 1825.
- [65] D. Nilsson, A. Jönsson, Design of evacuation systems for elevator evacuation in high-rise buildings, *J. Disaster Res.* 6 (6) (2011) 600–609.
- [66] E. Ronchi, D. Nilsson, Pedestrian movement in smoke: theory, data and modelling approaches, in: *In Crowd Dynamics*, uume 1, Springer, 2018, pp. 37–62.
- [67] E. Ronchi, “Testing the predictive capabilities of evacuation models for tunnel fire safety analysis,” *Saf. Sci.*, vol. 59, no. 0, pp. 141–153, Nov. 2013.
- [68] K. Fridolf, E. Ronchi, D. Nilsson, H. Frantzich, The representation of evacuation movement in smoke-filled underground transportation systems, *Tunn. Undergr. Space Technol.* 90 (Aug. 2019) 28–41.
- [69] E. Ronchi, S.M.V. Gwynne, D.A. Purser, P. Colonna, Representation of the impact of smoke on agent walking speeds in evacuation models, *Fire Technol.* 49 (2) (Apr. 2013) 411–431.
- [70] S. Gwynne, E.R. Galea, P.J. Lawrence, L. Filippidis, Modelling occupant interaction with fire conditions using the buildingEXODUS evacuation model, *Fire Saf. J.* 36 (4) (Jun. 2001) 327–357.
- [71] E. Ronchi, M. Kinsey, “Evacuation Models of the Future: Insights from an Online Survey on User’s Experiences and Needs,” Presented at the Advanced Research Workshop Evacuation and Human Behaviour in Emergency Situations EVAC11, 2011, pp. 145–155. Santander, Spain.
- [72] International Standards Organization, “Fire Safety Engineering – Assessment, Verification and Validation of Calculation Methods,” International Standards Organization, 2008. ISO 16730.
- [73] J. Lord, B. Meacham, A. Moore, R. Fahy, G. Proulx, Guide for Evaluating the Predictive Capabilities of Computer Egress Models NIST GCR 06-886, “National Institute of Standards and Technology, 2005.
- [74] Y. Wu, Testing the International Standards Organization Verification and Validation Protocol for Evacuation Simulations-An Application to the FDS+ Evac Model, “*LUTVDG/TVBB*”, 2019.
- [75] International Maritime Organization, Guidelines for Evacuation Analyses for New and Existing Passenger Ships, International Maritime Organization, 2007. MSC/Circ.1238.
- [76] International Maritime Organization, Revised Guidelines on Evacuation Analysis for New and Existing Passenger Ships, MSC.1/Circ.1533, International Maritime Organization, 2016.
- [77] G. Rimea, Richtlinie für Mikroskopische EntfluchtungsAnalysen [Guideline for Microscopic Evacuation Analysis] version 3.0, 2016. www.rimea.de.
- [78] E. Ronchi, E.D. Kuligowski, P.A. Reneke, R.D. Peacock, D. Nilsson, The Process of Verification and Validation of Building Fire Evacuation Models, “National Institute of Standards and Technology, 2013. Technical Note 1822.
- [79] E. Ronchi, E.D. Kuligowski, D. Nilsson, R.D. Peacock, P.A. Reneke, Assessing the verification and validation of building fire evacuation models, *Fire Technol.* 52 (Sep. 2014) 197–219.
- [80] R. Machado Tavares, E. Ronchi, “Uncertainties in Evacuation Modelling: Current Flaws and Future Improvements,” Presented at the 6th Human Behaviour in Fire Symposium 2015, Downing College, Cambridge, UK, 2015, pp. 185–196.
- [81] E. Smedberg, The Analysis of Results of Stochastic Evacuation Models, “*LUTVDG/TVBB*”, 2019.
- [82] R. Lovreglio, M. Spearpoint, M. Girault, The impact of sampling methods on evacuation model convergence and egress time, *Reliab. Eng. Syst. Saf.* 185 (2019) 24–34.
- [83] R. Lovreglio, E. Ronchi, D. Borri, The validation of evacuation simulation models through the analysis of behavioural uncertainty, *Reliab. Eng. Syst. Saf.* 131 (Nov. 2014) 166–174.
- [84] A. Grandison, S. Deere, P. Lawrence, E.R. Galea, The use of confidence intervals to determine convergence of the total evacuation time for stochastic evacuation models, *Ocean Eng.* 146 (Dec. 2017) 234–245.
- [85] A. Rigos, E. Mohlin, E. Ronchi, The cry wolf effect in evacuation: a game-theoretic approach, *Phys. Stat. Mech. Its Appl.* 526 (Jul. 2019) 120890.
- [86] P. Thompson, Integrating BIM and crowd evacuation dynamics: using a modern design platform directly to maintain data integrity and workflow. In *Pedestrian and Evacuation Dynamics 2018*, 2018. Lund, Sweden.
- [87] M. J. Seitz, A. Templeton, J. Drury, G. Köster, and A. Philippides, “Parsimony versus reductionism: how can crowd psychology be introduced into computer simulation?,” *Rev. Gen. Psychol.*, vol. 21, no. 1, pp. 95–102, Mar. 2017.
- [88] E.R. Galea, S. Deere, R. Brown, L. Filippidis, “An Evacuation Validation Data Set for Large Passenger Ships,” Presented at the Pedestrian and Evacuation Dynamics 2012 Conference, ETH, Zurich (Switzerland), 2012.